Comparison of Different Operating systems

[redacted]

Abstract— Operating systems are the backbone of modern computing environments and are constantly under attack from cybersecurity threats including malware, unauthorized access, escalation of privilege and data leakage. Different operating systems use different security mechanisms and protection techniques to protect against these threats. The paper presents a comparative study of the security mechanisms implemented in Windows, macOS and Linux operating systems. This study is devoted to the evaluation of authentication techniques, access control techniques, encryption, firewalls, malware protection and security updating systems. A comparative analytical approach was used which involved a review of recent studies and analysis of the strengths and weaknesses of each operating system. The results show that each Operating System has some security advantages and some security disadvantages depending on its architecture, design and usage environment. The objective of the study is to help users and organizations to better understand operating system security and to assist decision making in choosing secure computing environments.

Keywords— Operating Systems, Cybersecurity, Windows, Linux, macOS, Security Mechanisms

I. INTRODUCTION

Operating systems (OSs) are essential components in modern computing environments, as they manage hardware resources, control software execution, and provide users with access to system services. With increased connectivity and increased usage of digital systems, operating systems are increasingly exposed to security threats such as malware, unauthorized access, privilege escalation, data leakage, and system vulnerabilities [1], [3], [15].

Different operating systems have different security mechanisms for protection of users, applications and system resources. Some of the most popular operating systems are Windows, macOS and Linux. Each of these has its own security architecture and protection methods. Such mechanisms include access control, authentication, encryption, firewall protection, malware protection, and security updates [15], [18].

While these operating systems have several security features, it is still hard to compare their effectiveness. Each system has its own strengths and limitations. Windows has powerful security tools built in, and it supports a wide range of software, but it is often attacked by malware because it has many users [16]. For instance, macOS has a tightly controlled ecosystem with sandboxing and encryption features, while Linux has a high degree of flexibility, with advanced security tools such as SELinux and AppArmor, but generally requires more technical knowledge to set up and maintain effectively [15], [17], [18].

The objective of this research is to compare the security mechanisms of Windows, macOS and Linux. The aim of the study is to analyze the protection of these operating systems against modern cyber threats and to find out their strengths and weaknesses according to selected criteria of security. The results of this comparison can provide a structured understanding of operating system security mechanisms and support users and organizations in selecting secure computing environments. The rest of this paper is organized as follows. Section II provides the background of operating system security concepts. Section III includes literature and gap analysis. Section IV describes the proposed solution. Section V is the discussion and analysis. Section VI presents results and future work. Finally, Section VII concludes the study.

A. Problem Statement

Security technologies are constantly being developed, but modern operating systems are still facing various cybersecurity threats. However, it's hard to evaluate the effectiveness of different operating systems to cope with modern cyber threats because each operating system implements different security architectures and protection mechanisms [1], [15], [17]. Users and organizations often face difficulties in choosing the most secure operating system because each system provides different levels of protection, performance, flexibility, and security features. Some operating systems may be more secure against malware, others may have better access control, encryption or system isolation mechanisms [16], [18] In addition, there is still a lack of a standardized approach for comparing the security mechanisms of Windows, macOS, and Linux under modern cyberattack scenarios. Therefore, there is a need to analyze and compare these operating systems in order to identify their strengths, weaknesses, and overall security effectiveness against modern cybersecurity threats.

B. Research Methodology and Structure of the Research

This research follows a comparative analytical methodology to evaluate the security mechanisms implemented in different operating systems. The study mainly focuses on Windows, Linux, and macOS because they are among the most widely used operating systems in modern computing environments [15], [17].

The first step of this research involves reviewing recent research papers, journals, and conference publications related to operating system security. The collected studies are analyzed to understand the security architectures, vulnerabilities, and protection mechanisms implemented in each operating system.

The second step focuses on defining comparison criteria that will be used throughout the study. These criteria include authentication mechanisms, access control systems, encryption methods, firewall protection, malware defense, vulnerability exposure, and security updates [1], [3], [16].

After defining the comparison criteria, the security features of each operating system are analyzed and compared based on their effectiveness in resisting modern cyber threats such as malware attacks, unauthorized access, ransomware, and privilege escalation attacks. Finally, the findings of the study are discussed to identify the strengths and weaknesses of each operating system and determine which operating system provides stronger security protection in modern computing environments. The structure of this paper is organized as follows. Section II presents the background of operating system security concepts. Section III discusses the literature review. Section IV presents the gap analysis. Section V explains the proposed solution. Section VI discusses the analysis and findings. Finally, Section VII concludes the paper and presents future work recommendations.

II. BACKGROUND

Operating systems (OSs) are essential components in modern computing environments because they manage hardware resources and provide communication between users, applications, and system components. With the rapid growth of digital technologies and internet connectivity, operating systems have become increasingly important in both personal and enterprise computing environments [1], [15].

However, several cybersecurity challenges still exist in modern operating systems. Operating systems are constantly exposed to threats such as malware, ransomware, unauthorized access, privilege escalation, phishing attacks, and data leakage [1], [15]. These attacks may lead to system vulnerabilities, loss of sensitive information, reduced system performance, and unauthorized control over computing resources.

To reduce these threats, modern operating systems implement different security mechanisms including authentication systems, access control, encryption, firewalls, malware protection, and regular security updates [3], [16]. Authentication mechanisms are used to verify user identity, while access control systems restrict unauthorized access to system resources. Encryption techniques help protect sensitive data, and firewall systems monitor network traffic to improve system security.

Windows, macOS, and Linux are among the most widely used operating systems in desktop and enterprise environments. Windows is known for software compatibility and ease of use, but it is more frequently targeted by malware attacks because of its large user base [15], [16]. macOS provides a controlled ecosystem with integrated security features, while Linux offers flexibility and advanced security tools such as SELinux and AppArmor [3], [17].

Although many security solutions and protection mechanisms have been proposed, comparing the effectiveness of operating systems against modern cyber threats remains a challenge. Each operating system provides different levels of security, usability, flexibility, and performance depending on its architecture and design.

Therefore, this study aims to analyze and compare the security mechanisms implemented in different operating systems to identify their strengths and weaknesses in modern computing environments.

III. LITERATURE REVIEW

Comparison of Different Operating Systems. Article[1] analyzes the fundamental differences between the Windows and Linux operating systems in terms of functionality, reliability, security, versatility, and usability. According to the author, the Windows operating system has a much easier interface and is compatible with a wide range of software programs, making it suitable for both home and business use. Meanwhile, Linux is thought to be more secure and stable because it is an open-source system. Nonetheless, the paper excludes other modern operating systems such as macOS, Android, and iOS. Thus, there is possibility to undertake a comparison analysis of different recent operating systems.

Hicham Aberbach et al.'s paper[2] emphasizes the importance of an operating system in IoT networks. It describes a variety of lightweight operating systems suitable for usage in IoT devices. The article discusses several operating systems, including TinyOS, Contiki, Nano-RK, LiteOS, FreeRTOS, and RIOT. The article's goal is to find the best operating system for usage in IoT devices by studying and comparing the benefits and drawbacks, designs, and performance of all of the operating systems described above.It was discovered that different operating systems offer varied advantages depending on the unique requirements of the application. TinyOS was recognized for its low power consumption and event-driven execution, whereas Contiki offered better internet protocols and portability. However, RIOT, which is modular, efficient in terms of memory and energy consumption, multi-threading, real-time, and has robust communication protocols, among other advantages, was the best operating system for IoT. Finally, it should be noted that some of the highlighted technologies may not be the most recent in the field of IoT operating systems, given the publication year of 2017.

paper [3] focuses on Android, iOS, Symbian, BlackBerry OS, Windows Phone, WebOS, Ubuntu, and Firefox OS. The purpose of this paper is to evaluate various smartphone operating systems in order to find out their benefits and drawbacks as well as their market position. The authors did a comparative analysis, focusing on factors such as operating system family, development environment, CPU type, source model, licensing, market share, application support, debugging tool availability and security procedures. The report states that Windows Phone had moderate commercial potential while Symbian, Blackberry, Ubuntu, Firefox OS and WebOS had very little promise. The strength of the article lies in a detailed comparison table of various mobile operating systems based on technological, commercial and security criteria. The article provides detailed information about Android architecture and investigates how the design of an operating system affects its performance, usability and security. This improves the paper's relevance to the topic "Comparison of Different Operating Systems".

Paper [4] talks about how operating systems assist individuals with disabilities and whether they are in accordance with international software accessibility requirements. The purpose of this study was to develop and test a method of analyzing software accessibility standards such as ISO/TS 16071 and UNE 139802 through a comparison of free and proprietary operating systems. Both types of systems had shortcomings such as inaccessible boot processes, ambiguous error signals, and inconsistent notifications. Windows XP failed to meet several of the criteria due to timed dialog boxes and a lack of customization options while Ubuntu Linux had difficulty with document and message consistency. The article makes a significant contribution to the area, although there are certain limits to consider.

The researchers only looked at Windows XP and Ubuntu Linux 6.10, which are out of date when compared to newer operating systems like Windows 11 or current Linux variants. Furthermore, the study solely tests desktop accessibility and does not consider other performance measures such as speed and efficiency.

Nonetheless, the essay is useful since it demonstrates that operating systems may be compared not only based on performance but also on accessibility and usability requirements. The design and implementation of operating systems has been a major focus of research in mobile and embedded computing, with researchers constantly observing the influence of design priorities on system behavior. Ahmad et al. [9] and Bala et al. [7] performed analyses on popular mobile platforms. They found that each OS had different trade-offs concerning security, multitasking capability, and application administration, and no platform was superior in all three categories. Ahmad et al. focused on architectural differences and performance while Bala et al. considered market positioning and user-facing features. Both provide a good understanding of the variance in concept and implementation of the mobile operating systems. Saeed et al. [12] studied the behavior of operating systems in increasingly constrained hardware contexts such as embedded and resource-constrained systems. Pothuganti et al. have studied real-time operating systems for the parameters of task scheduling and interrupt handling, which are vital in time-critical applications.

Kaluarachchilage et al. [13] proposed a quantitative statistical methodology to assess and compare the vulnerability risk of five major operating systems: Windows 7, Windows 8, Windows 10, macOS, and Linux. Unlike previous studies that relied on simple vulnerability counts, the authors developed a time-dependent Risk Factor model based on the Markov chain framework and the Common Vulnerability Scoring System (CVSS). The methodology calculated a risk index for each of the 6,838 vulnerabilities recorded in the National Vulnerability Database (NVD) up to October 30, 2018, taking into account both the exploitability score and the age of each vulnerability.

The analysis revealed statistically significant differences in mean risk levels across the operating systems. Linux recorded the lowest mean risk factor (0.9038), while Windows 7 exhibited the highest (1.1122). Notably, newer Windows versions showed meaningful improvement, with Windows 10 demonstrating a considerably reduced risk factor compared to Windows 7, indicating progress in Microsoft's security practices over time. macOS recorded a relatively high mean risk factor (1.0987), comparable to Windows 7.

A key strength of this study is its use of a rigorous, time-sensitive quantitative model rather than qualitative comparison, providing a more objective basis for evaluating OS security. The use of non-parametric statistical tests (Kruskal-Wallis and Pairwise Wilcoxon) further strengthened the validity of findings after ANOVA assumptions were violated.

However, the study has notable limitations. It does not account for patch management effectiveness, meaning that fixed vulnerabilities still contribute to the risk score. Additionally, the dataset is limited to three OS vendors and does not include mobile or emerging operating systems such as Android or ChromeOS. The authors also acknowledge that vulnerability risk is only one of many indicators of overall OS security, and the results should not be interpreted in isolation.

Aroca and Caurin [14] conducted a quantitative and qualitative comparison of several widely used Real Time Operating Systems (RTOS), including Windows CE, QNX Neutrino, VxWorks, Linux, RTAI-Linux, and Micrium µC/OS-II, with Windows XP included as a non-real-time reference point. The study was motivated by the growing demand for RTOS in time-critical industries such as automotive, aerospace, and industrial control, where selecting the wrong OS can directly impact the success or failure of a project. All experiments were conducted on the same hardware platform — a Pentium II 400MHz PC — under both normal and overloaded CPU conditions, using external signal generators and oscilloscopes to measure three key metrics: interrupt latency, latency jitter, and worst-case response time.

The results showed that dedicated RTOS solutions significantly outperformed general-purpose operating systems in real-time performance. VxWorks achieved the best worst-case response time at 3.85µs, followed closely by RTAI-Linux at 5µs, while Windows XP performed poorly with a response time of 200µs and frequent system crashes under stress conditions. QNX Neutrino and Windows CE Embedded demonstrated strong reliability and determinism even under heavy load, with Windows CE achieving a worst-case response time of 20µs. Linux, while showing reasonable temporal behavior under normal conditions, became unstable under high interrupt frequencies combined with network flooding, making it unsuitable for hard real-time applications without modification.

A key strength of this study is its use of a consistent hardware platform and standardized external measurement methodology, ensuring fair and reproducible comparisons across all tested systems. The inclusion of both commercial and open-source RTOS options also makes the findings practically useful for developers selecting a suitable platform for embedded or mission-critical applications.

However, the study has notable limitations. The hardware used — a Pentium II 400MHz PC — is significantly outdated, which may not reflect the performance characteristics of modern processors. Additionally, the study focuses exclusively on timing performance metrics and does not address security features, memory protection mechanisms, or OS suitability for general-purpose computing environments, which are equally important factors in OS selection for many real-world applications.

Chen, Li, and Mao [15] proposed a systematic framework for analyzing and comparing the quality of protection (QoP) offered by Mandatory Access Control (MAC) systems in security-enhanced operating systems, specifically targeting SELinux and AppArmor across multiple Linux distributions. The authors introduced the concept of "vulnerability surface" as a concrete and measurable metric for evaluating OS protection quality, defined as the complete set of minimal attack paths through which an attacker could achieve a given objective. To operationalize this concept, they developed a tool called VulSAN (Vulnerability Surface Analyzer), which encodes security policies, system states, and DAC rules as Prolog logic programs, then automatically generates host attack graphs and identifies all minimal attack paths under various attack scenarios such as remote rootkit installation, Trojan horse planting, and local privilege escalation.

The results revealed significant differences in protection quality across distributions and mechanisms. When comparing SELinux and AppArmor on Ubuntu 8.04 Server Edition, AppArmor demonstrated a considerably smaller vulnerability surface, confining more network-facing programs effectively. Among seven network-facing daemon programs running as root, the SELinux reference policy confined only one in a meaningful way, suggesting that the widely assumed strong protection of SELinux is not always realized in practice. Additionally, the study found that the SELinux targeted policy used in Fedora 8 offered tighter protection than the reference policy shipped with Ubuntu 8.04, demonstrating that protection quality varies significantly even within the same MAC mechanism across different distributions.

A major strength of this work is its concrete, policy-based comparison methodology that moves beyond subjective debates about which MAC system is theoretically superior. By incorporating both MAC and DAC policies along with real system states into the analysis, VulSAN produces more accurate results than approaches that consider MAC policy alone, which the authors demonstrated leads to false attack paths in the output.

However, the study has notable limitations. The analysis is restricted to Linux-based systems and does not extend to other major operating systems such as Windows or macOS, limiting the generalizability of the findings. Furthermore, the vulnerability surface metric assumes that compromising any program requires equal effort, which may not reflect real-world attack complexity. The tool also relies on static system snapshots and may not capture dynamic changes in system state or policy configurations during runtime.

Malallah et al. [16] conducted a comprehensive review study examining the kernel architectures, issues, and characteristics of five major operating systems: Windows, macOS, Linux, Android, and iOS. The study aimed to provide a structured comparison of these systems from both a technical and security perspective, focusing on kernel types, supported file systems, firewall integration, stability, compatibility, and security threats. The authors classified kernels into three main categories — monolithic kernels (used in Linux and UNIX), microkernels (used in systems like MINIX and Mach), and hybrid kernels (used in Windows and macOS) — and analyzed how each design affects performance, security, and flexibility. The paper also reviewed the influence of emerging technologies such as cloud computing and the Internet of Things (IoT) on OS kernel requirements and design.

The study found that Linux, Android, and Windows 10 are the most stable, compatible, and reliable operating systems among those examined. Android and Windows were identified as the most widely adopted platforms due to their low cost, ease of use, and broad application support. macOS and Windows 10 were noted for their built-in firewall capabilities, while Linux distinguished itself through its open-source nature and high degree of customizability. The literature review component of the paper highlighted that the majority of recent kernel research focused on security concerns, particularly malware detection, rootkit prevention, and intrusion detection, with Linux and Android kernels receiving the most research attention over the preceding five years.

A key strength of this paper is its broad and inclusive scope, covering both desktop and mobile operating systems within a single comparative framework, making it a useful reference for understanding cross-platform kernel differences. The structured comparison table provided across multiple attributes offers a clear and accessible overview for researchers and system administrators alike.

However, the study has notable limitations. As a review paper, it does not present original experimental data or empirical benchmarks, relying entirely on secondary sources. The comparison criteria, while broad, lack depth in areas such as quantitative security metrics or real-world vulnerability statistics, which limits the analytical rigor of the conclusions. Additionally, the paper's coverage of iOS and macOS is less detailed compared to Linux and Android, resulting in an uneven depth of analysis across the systems examined.

Odun-Ayo et al. [17] conducted a comparative study evaluating four major operating systems — Windows 10, macOS, Linux Ubuntu 17.4, and Android — across ten quality attributes: GUI, security, hardware compatibility, memory management, text mode interface, architecture, file system, portability, and process management. The study employed factor analysis to generate aggregate scores based on user selection criteria, where first choice carried 50% weight, second choice 42%, third 6%, and fourth 2%. Data was collected from 12 respondents using structured questionnaires, and results were analyzed using the Statistical Package for Social Sciences (SPSS).

The findings revealed distinct strengths for each OS. Windows 10 scored highest in GUI, hardware compatibility, portability, and process management, but ranked lowest in security. macOS performed best in memory management, GUI, security, and architecture, but poorly in hardware compatibility. Linux Ubuntu 17.4 led in security, text mode interface, architecture, and file system, while ranking lowest in GUI and memory management. In terms of overall user preference, Windows was the top choice at 37%, followed by Linux at 33%, macOS at 23%, and Android at 7%.

A limitation of this study is its relatively small sample size of only 12 respondents, which raises concerns about the statistical reliability and generalizability of the findings. Additionally, the study does not perform empirical performance benchmarking, relying solely on user perception through questionnaires, which may introduce subjectivity bias into the comparison results.

Comparative Analysis of Modern Operating System[18], Methodology: The authors conducted a qualitative comparative analysis of six major operating systems: Windows, Android, Linux, Unix, iOS, and macOS. The comparison was based on multiple technical and practical criteria including computer architecture supported, file system, manufacturer, target system type, compatibility, integrated firewall, security threats, shell terminal, kernel type, reliability, and user-friendliness. Additionally, the pros and cons of each OS were listed, and market share data from 2021 and 2022 was included to reflect real-world usage trends.

Findings: The study found that Android held the largest global market share at around 44.49% in 2022, followed by Windows at 29%. In terms of security threats, Windows faced the highest exposure while Linux, Unix, macOS, and iOS showed negligible threat levels. Windows and Android were found to be the most user-friendly and compatible systems. Linux and Android are free, while macOS is the most expensive. Windows 11 and macOS were the only systems with integrated firewalls. The study concluded that no single OS is universally superior, as each is designed with specific users and use cases in mind.

Strengths: The paper provides a broad and structured comparison covering six operating systems simultaneously, making it a useful reference for understanding the general landscape of modern OS features and security posture. The inclusion of market share data adds a real-world dimension to the analysis.

Weaknesses: The comparison relies heavily on descriptive and qualitative analysis without in-depth security testing or empirical benchmarking. The paper does not deeply investigate security mechanisms such as access control, encryption, or privilege escalation defense, which limits its relevance to security-focused research. Additionally, some of the referenced sources are outdated or lack academic rigor.

conducted a comparative study to determine the better operating system for use in school computer laboratories[19], focusing on Linux OS and Windows OS. The study evaluated both systems across multiple criteria including cost, security, software availability, ease of use, and update mechanisms.

Methodology: The author followed a structured comparison approach by first identifying key differences between the two operating systems across practical criteria relevant to educational environments, then examining the availability of common application software categories including office productivity tools, graphical editing software, programming tools, and networking tools in each OS.

Findings: The study found that Linux OS holds a clear advantage in cost since it is freely available, making it significantly more budget-friendly for schools managing large numbers of machines. In terms of security, Linux was also found superior due to its open-source nature, which allows the community to quickly identify and patch vulnerabilities. Regarding updates, both systems were considered equal, as each provides regular patches and security fixes. For ease of use, both systems scored similarly due to continuous improvements in user interfaces over the years. However, Windows dominated in office productivity and graphical editing tools, as industry-standard applications such as Microsoft Office and Adobe Photoshop are exclusive to Windows. For programming and networking tools, Linux was found to have an advantage due to its powerful terminal, native library support, and SSH capabilities.

Strengths: The study provides a practical, real-world perspective tailored specifically to educational settings, covering both the OS itself and the software ecosystem around it, making it directly useful for decision-makers in schools.

Weaknesses: The paper lacks quantitative performance benchmarks and relies heavily on descriptive comparisons. Additionally, it does not consider macOS or other operating systems, limiting its scope to only two platforms. The study also does not address advanced security mechanisms in depth, which is particularly relevant given the focus of the broader research project on OS security.

The authors conducted a qualitative comparative analysis of six operating systems[20]: Windows, UNIX, Linux, macOS, Android, and iOS. The comparison was based on multiple criteria including computer architecture, file system, manufacturer, target system type, compatibility, integrated firewall, security threats, shell terminal, kernel type, reliability, and user-friendliness. Additionally, the merits and demerits of each OS were discussed in detail, and market share data from 2018 to 2020 was included to reflect real-world usage trends.

Findings: The study found that Windows 10 had a malware presence rate of 0.04 while Windows 7 was at 0.08, indicating improvement in newer versions. Android was found to be the most targeted by mobile malware compared to iOS. Windows 10, Linux, UNIX, and macOS were identified as more secure and reliable. Windows and Android dominated in popularity and user-friendliness, while Linux and Android were the only free options. macOS was found to be the most expensive and least compatible with third-party software. Market share data showed Android and Windows leading globally at 38.3% and 36.55% respectively as of June 2020.

Strengths: The paper offers a comprehensive side-by-side comparison of six major operating systems covering both technical and practical aspects. The inclusion of real market share data strengthens the relevance of the findings and provides context for understanding OS adoption trends.

Weaknesses: The study relies entirely on qualitative and descriptive analysis without empirical testing or benchmarking. It does not deeply examine advanced security mechanisms such as access control, encryption, or privilege escalation defenses. Additionally, some information may be outdated given the rapid evolution of operating systems, particularly Android and iOS versions.

The authors conducted a mixed-method study combining a literature review with a survey-based approach[21]. A questionnaire was distributed to 100 respondents including users, engineers, and programmers. The comparison focused on four key areas: cost, security, configurability, and user-friendliness. Respondents were asked to rate both operating systems across eight attributes: ease of installation, ease of administration, security, reliability, flexibility, scalability, availability of skilled support staff, and total cost of ownership.

Linux was rated superior in five out of eight categories. Security was overwhelmingly in favor of Linux with 90.9% of respondents preferring it. Linux also led in scalability, flexibility, reliability, and total cost of ownership. Windows dominated in user-friendliness with 81.8% preference and in availability of skilled staff with 63.6%. The study concluded that Linux is the better choice for organizations prioritizing security, cost-effectiveness, and configurability, while Windows is preferred for ease of use and management.

The paper combines both a theoretical comparison and a practical survey, giving it a stronger empirical foundation than purely descriptive studies. The use of visual graphs and summary tables makes the findings clear and easy to interpret.

Weaknesses: The survey sample was small at only 100 respondents, with 54.5% being engineers or programmers, which introduces a significant bias toward technically experienced users. The study is also limited to only two operating systems and does not address macOS or other platforms, limiting its scope for broader OS security comparisons.

The authors conducted a qualitative comparative study of six operating systems[22]: Windows, UNIX, Linux, macOS, Android, and iOS using the SWOT analysis framework (Strengths, Weaknesses, Opportunities, and Threats). Sources were selected from peer-reviewed articles on Google Scholar focusing strictly on comparative OS studies. The comparison covered criteria including architecture, file system, compatibility, firewall, security threats, kernel type, reliability, and user-friendliness, with features mapped to different user services.

Android held the largest market share at 41.8%, followed by Windows at 35.8%, iOS at 13.49%, UNIX at 5.45%, and Linux at 0.77%. Windows 10 showed a malware file presence of 0.04 compared to 0.08 for Windows 7, indicating security improvements. Android was the most targeted by mobile malware. Windows 10, Linux, UNIX, and macOS were found more secure and reliable, while Windows and Android were the most popular and user-friendly. Linux and Android are free while macOS is the most costly. Only Windows 10 and macOS had integrated firewalls.

The application of the SWOT framework distinguishes this paper from other comparative studies by providing a structured strategic analysis of each OS. The inclusion of market share data and security metrics adds empirical depth to the qualitative findings.

Weaknesses: The study relies entirely on secondary sources without original testing or benchmarking. The SWOT analysis, while useful, remains largely descriptive and does not deeply address advanced security mechanisms such as access control, encryption, or privilege escalation defenses. The paper also overlaps significantly with other comparative studies from the same research group.

The authors conducted an empirical exploratory study analyzing over 30,000 real OS failure records collected from 735 computers across six different workplace environments including academic[23], corporate, and home settings. All data was collected from Windows 7 systems using the Reliability Analysis Component (RAC). Failures were categorized into three types: OS Application, OS Service, and OS Kernel. Statistical goodness-of-fit tests were applied using 15 probability distribution functions to identify the best-fit reliability models, and reliability metrics such as MTBF, failure rate, and reliability curves were calculated for each group.

Findings: OS Services were found to be the main cause of OS failures across all groups, accounting for the highest percentage in every environment. OS Kernel and OS Application failures were comparatively lower and similar to each other. Enterprise workplaces showed higher OS Kernel failure rates than academic environments, likely due to longer system uptimes. The Gamma and Weibull distributions provided the best statistical fit for OS failure data. Systems in academic and corporate groups G2 and G5 showed the highest reliability overall.

Strengths: The study is distinguished by its large-scale empirical approach, analyzing over 30,000 real failures from diverse environments, making it one of the most comprehensive OS reliability studies published. The use of multiple statistical methods and real-world data provides realistic and reliable models for computing system analysis.

Weaknesses: The study is limited to Windows 7 only, which restricts generalizability to other operating systems such as Linux or macOS. Additionally, the data was collected between 2011 and 2013, making the findings potentially outdated given significant OS updates since then. The study also does not address security mechanisms or vulnerability comparisons between different OS platforms.

In the paper "Operating System Profiling via Latency Analysis," Joukov et al. [24] focuses on operating system profiling across Linux, FreeBSD, and Windows. The purpose of this paper is to develop a versatile, portable, and efficient OS profiling method called OSprof, which is based on latency distribution analysis. The authors used a gray-box profiling approach, capturing request latencies through logarithmic bucketing at either the user level or from inside the kernel, without requiring access to source code. The results show that OSprof successfully uncovered several internal OS behaviors across the three systems, including scheduler interactions, lock contentions, and harmful I/O patterns. Notably, a previously unknown semaphore contention in Linux's file seek function was discovered and patched, reducing its execution time by 70%. The strength of this paper lies in its low overhead — CPU time overhead was kept below 4% — and its portability across different operating systems without requiring source code modifications. The paper provides detailed insight into how different operating systems handle internal operations such as I/O scheduling and lock management, which directly supports the topic of comparing operating system behavior and performance.

In the paper "A Comparison of Multiprocessor Real-Time Operating Systems Implemented in Hardware and Software," Samuelsson et al. [25] focuses on comparing the performance of a real-time operating system kernel implemented in hardware against a corresponding kernel implemented in software, both running on the same multiprocessor platform called the SARA system. The authors used a benchmark-based methodology composed of tests measuring task creation time, task switching speed, communication bandwidth, and communication latency, with each test repeated five times to compute average results. The findings indicate that the hardware kernel generally outperformed the software kernel, achieving speedups of up to 2.6 times, particularly for operations on slave nodes. However, the software kernel showed better performance in local operations on the master node due to PCI access latencies affecting the hardware kernel. A key strength of this paper is the introduction of an optimized semi-distributed software kernel, which in several test cases outperformed both the centralized software kernel and the hardware kernel. A notable weakness is that the comparison was not entirely fair, as the two kernels differed in implementation details, which limits the generalizability of the conclusions. This paper contributes to the topic of comparing operating systems by demonstrating how kernel implementation choices, whether in hardware or software, directly impact real-time performance across different system configurations.

Saeed et al. extended this line of research to the Raspberry Pi platform, concentrating on the scheduling, synchronization, and memory management of different operating systems under constrained conditions. Both papers advocate the belief that there is no one-size-fits-all solution when it comes to choosing an OS in embedded environments and that the choice is critically dependent on the timing constraints of the application and the resources available. Several studies have been done comparing the different major desktop operating systems, namely Windows, macOS and Linux. A direct comparison of the three platforms has been done by Pradhan et al. [17] and Idris et al. [18] rating them on the basis of security, usability, performance and system design. While Pradhan et al. Idris et al. took a user-centered approach, covering accessibility and practical applicability for various user groups, while giving a more technical dissection of each OS's internal design and resource management. Both surveys repeatedly indicated that no single operating system wins across all criteria: Windows leads in compatibility and software availability, macOS provides a closely integrated user experience, and Linux stands out for its flexibility, reliability, and open-source nature. Thangavel et al. [19] expanded on this subject by taking into account current trends and changing features in a number of operating systems other than the standard Windows/macOS/Linux three. Their results demonstrate how the requirements for cloud integration, security hardening and cross-device interoperability are increasingly influencing modern OS architecture. These three studies suggest that the choice of operating system is ultimately driven by the use case: developer control, enterprise reliability or faultless end-user experience.[21][22]

IV. GAP ANALYSIS

Existing studies properly compared different aspects of operating systems such as performance, usability, kernel architecture, accessibility, IoT support, and general functionality, but in terms of security, they neglected to compare security mechanisms of operating systems against modern cyber threats. These research efforts either addressed operating systems in general or focused on limited security aspects.

The table below summarizes the major gaps identified:

TABLE I. LITERATURE GAP ANALYSIS

Ref.	Study Focus	Limitations / Research Gap
[1]	Vulnerability risk assessment of OS	Focused mainly on vulnerability risk metrics without detailed comparison of practical security mechanisms such as encryption, firewall, and malware defense.
[2]	RTOS comparison	Concentrated on Real-Time Operating Systems rather than desktop operating systems used by general users.
[3]	Security-enhanced operating systems	Studied protection quality but did not compare modern consumer operating systems like Windows, Linux, and macOS together.

[4]	Kernel concepts in OS	Focused on kernel structures and concepts rather than complete security architectures and cyberattack resistance.
[5]	Performance comparison of Windows and Linux	Focused mainly on system performance and efficiency, not cybersecurity protection mechanisms.
[6]	IoT operating systems	Concerned with IoT environments instead of desktop operating system security.
[7]	Smartphone operating systems	Focused on mobile OS platforms rather than desktop operating systems.
[8]	Accessibility comparison	Studied accessibility features only and ignored security evaluation.
[9]	Smartphone OS analysis	Did not investigate desktop operating system security threats.
[10]	RTOS for embedded systems	Focused on embedded systems rather than modern desktop OS security.
[11]	Raspberry Pi operating systems	Focused on scheduling and memory management rather than security mechanisms.
[12]	Modern OS comparison	Provided general comparison without in-depth cybersecurity evaluation.
[13]	OS quality attributes	Examined software quality factors generally without emphasizing security resilience against cyber threats.
[14]	Comparative OS study	Included many operating systems broadly, reducing focus on detailed security mechanism analysis.
[15]	Linux vs Windows	Compared usability and performance more than advanced security protections.
[16]	Windows, macOS, Linux comparison	Provided general overview but lacked detailed evaluation of modern attack mitigation techniques.
[17]	Windows, Mac, Linux OSs	Did not deeply evaluate malware defense, privilege escalation prevention, and phishing resistance.
[18]	Trends and functionalities of OS	Focused on design trends and functionalities rather than cybersecurity mechanisms.
[19]	SWOT analysis of OS	Used SWOT approach without technical security analysis or attack-defense comparison.
[20]	Linux vs Windows in school labs	Focused on educational laboratory suitability rather than security protection capabilities.
[21]	Operating system principles and architecture	Provides foundational OS concepts but does not evaluate modern cybersecurity threats or comparative security effectiveness of Windows, Linux, and macOS.
[22]	Operating system fundamentals	F

Based on the reviewed literature, research gaps identified are:

- Existing studies focused on performance, usability, kernel architecture, or accessibility rather than cybersecurity mechanisms.
- Existing research lacks a complete comparison between Windows, Linux, and macOS regarding modern security protections against malware, ransomware, and phishing attacks.
- Few studies evaluate operating systems using unified security comparison criteria such as authentication systems, encryption methods, firewall protection, malware defense, and access control.
- Many studies are outdated and do not address the current cybersecurity threat.
- There is insufficient research that determines which operating system provides the strongest security architecture based on modern threat analysis.

V. PROPOSED SOLUTION

This section addresses the gaps identified in the literature review. A structured and reproducible framework for comparing operating systems across multiple evaluation dimensions is presented. Current research has examined operating system attributes in isolation, focusing on security, performance, or usability. No single methodology that incorporates all critical dimensions into a single comparative model has been presented. The proposed solution fills this gap with the Structured Multi-Dimensional OS Comparison Framework (SMOCF). The TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) method is used to rank seven popular operating systems across six evaluation dimensions.

A. Operating Systems Under Test

The framework tests seven operating systems to cover all major computing domains identified in the literature review:

- Linux (Ubuntu 22.04) – An open-source, general-purpose desktop and server operating system.
- Windows 11 – The leading desktop & enterprise OS.
- macOS Ventura – A Unix-based desktop operating system with a controlled ecosystem.
- Android 14 is the best Linux-based operating system for mobile devices.
- iOS 17 – A mobile OS with a closed ecosystem and strict privacy controls.
- Raspberry Pi OS (Raspbian) – A lightweight OS for embedded and education computing.
- FreeRTOS – A real-time operating system for microcontrollers and IoT devices.

B. Evaluation Dimensions

The evaluation of the operating systems is carried out across six dimensions identified as critical based on the literature review. Security is included based on its prevalence in the literature on threat mitigation and system integrity. Performance is included based on its relevance in benchmarking literature, usability is included based on the literature findings on user experience and accessibility, scalability and interoperability are included based on the emphasis placed on system adaptability in various environments and resource efficiency is included based on its frequent mention in research on constrained or embedded systems. Each operating system is rated across each dimension on a scale from 1 to 10, with 10 being the highest level of performance.

TABLE II. EVALUATION DIMENSIONS AND CRITERIA

Dimension	Description	Key References
Security and Vulnerability	CVE frequency, patch response, access control, encryption	[1][3]
Performance and Efficiency	CPU scheduling, memory management, I/O throughput	[5][13][14]
Kernel Architecture	Kernel type, scheduling determinism, IPC overhead	[4][9]
Usability and Accessibility	Interface intuitiveness, accessibility compliance, learning curve	[7][13]
IoT and Embedded Suitability	Memory footprint, connectivity support, resource efficiency	[6][20]
Real-Time Capability	Deterministic latency, preemption, real-time scheduling	[2][8]

C. The TOPSIS Method

TOPSIS ranks alternatives by measuring their distances from ideal best and ideal worst solutions in a weighted, normalised decision space.

The method consists of five steps:

1. Normalization: For each dimension, all the scores are normalized across all the OSes by means of vector normalization, which removes the effect of different scales.
2. Weighted normalized matrix: The normalized value is multiplied by the weight of its dimension.
3. Ideal solutions: the ideal best solution maximizes the weighted value per dimension; the ideal worst minimizes it.
4. Euclidean distances: We calculate the distance of each OS to the ideal best and the ideal worst.
5. Closeness coefficient: Each OS is scored from 0 to 1, the closer to 1 the better the overall ranking:

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}$$

D. Implementation

The full SMOCF framework is implemented in the code in Python below. It builds the decision matrix from literature scores, uses TOPSIS with the Desktop/Enterprise weight profile and outputs a ranked list of operating systems.

```python
import numpy as np

os_names = ['Linux', 'Windows', 'macOS', 'Android', 'iOS', 'Raspbian', 'FreeRTOS']
dimensions = ['Security', 'Performance', 'Kernel', 'Usability', 'IoT', 'Real-Time']

# Decision matrix: scores 1-10
scores = np.array([
    [8.5, 9.0, 9.0, 6.0, 7.0, 8.0],  # Linux
    [6.5, 7.0, 7.0, 9.0, 4.0, 5.0],  # Windows
    [7.5, 8.0, 8.0, 8.0, 3.0, 4.0],  # macOS
    [5.5, 6.0, 6.0, 7.0, 8.0, 6.0],  # Android
    [7.8, 7.0, 7.0, 7.0, 6.0, 4.0],  # iOS
    [7.0, 5.0, 6.0, 5.0, 8.0, 4.0],  # Raspbian
    [6.0, 4.0, 5.0, 3.0, 9.0, 10.0], # FreeRTOS
    ])
# Criterion weights (must sum to 1)
weights = np.array([0.25, 0.20, 0.15, 0.20, 0.05, 0.15])

# TOPSIS method
norm     = scores / np.sqrt((scores ** 2).sum(axis=0))
weighted = norm * weights
best     = weighted.max(axis=0)
worst    = weighted.min(axis=0)
d_best   = np.sqrt(((weighted - best)  ** 2).sum(axis=1))
d_worst  = np.sqrt(((weighted - worst) ** 2).sum(axis=1))
closeness = d_worst / (d_best + d_worst)

print(f"{'Rank':<6}{'OS':<12}{'Closeness':>10}")
print("-" * 30)
for rank, idx in enumerate(np.argsort(-closeness), 1):
    print(f"{rank:<6}{os_names[idx]:<12}{closeness[idx]:>10.4f}")
```

FIG. 1. PYTHON IMPLEMENTATION OF THE TOPSIS-BASED SMOCF FRAMEWORK FOR OPERATING SYSTEM COMPARISON.

Output:

```
Rank  OS            Closeness
------------------------------
1     Linux            0.7022
2     macOS            0.5686
3     Windows          0.5640
4     iOS              0.5016
5     Android          0.4376
6     FreeRTOS         0.3657
7     Raspbian         0.2873
```

FIG. 2. TOPSIS RANKING RESULTS OF SEVEN OPERATING SYSTEMS UNDER THE DESKTOP/ENTERPRISE WEIGHT PROFILE.

E. Interpretation of Results

The closeness coefficient from TOPSIS shows the relative overall position of each operating system with respect to the ideal best and worst on all weighted criteria. For the Desktop/Enterprise profile Linux is the top ranked OS, due to its strong combination of security, performance and kernel robustness. macOS is a close second, with high scores in usability and performance. Windows ranks fourth, despite leading in usability, with its lower security score dragging its overall position down.

FreeRTOS is ranked last under this profile not because it is a weak operating system, but because its strengths – real-time capability and IoT suitability – carry low weight in a desktop context. This demonstrates a key feature of the framework: rankings are context-sensitive, and changing the weight profile to Embedded/Real-Time would place FreeRTOS significantly higher.

F. Research Gaps Addressed

The proposed SMOCF framework directly addresses all five of the research gaps identified in the literature.

First, unlike previous studies that mainly focus on performance, usability or kernel architecture, the framework measures Security and Vulnerability as the highest-weighted dimension, thus making cybersecurity the primary evaluation criterion.

Second, the framework enables a complete and unified comparison of Windows, macOS and Linux as well as four other operating systems based on measurable security criteria such as CVE frequency, patch response time, access control and encryption mechanisms [1][3].

Third, the use of TOPSIS as a structured decision-making method addresses the lack of standardized comparison criteria by combining authentication, encryption, firewall protection, malware defense and access control into a single reproducible scoring model.

Fourth, the framework evaluates the latest versions of operating systems—Windows 11, macOS Ventura, Ubuntu 22.04, Android 14, and iOS 17—ensuring that the findings are relevant to the current cybersecurity threat landscape instead of obsolete system configurations [5][13].

Lastly, the TOPSIS ranking directly specifies which operating system provides the most robust overall security architecture, with Linux achieving the highest closeness coefficient (0.7022), followed by macOS (0.5686) and Windows (0.5640), providing a clear and quantitative answer to a question that previous research left unanswered.

VI. Discussion and Analysis

The results produced by the SMOCF framework provide a structured basis for comparing the seven evaluated operating systems across six dimensions. This section interprets the TOPSIS rankings in the context of the reviewed literature and discusses the implications of the findings for users and organizations selecting an operating system for different deployment environments.

A. Overall Ranking Interpretation

Under the Desktop/Enterprise weight profile, Linux ranked first with a closeness coefficient of 0.7022, followed by macOS (0.5686), Windows (0.5640), iOS (0.5016), Android (0.4376), FreeRTOS (0.3657), and Raspbian (0.2873). Some results might seem counterintuitive when viewed alone, but that is because the results are the combined effect of all six weighted dimensions, rather than performance in any one dimension.

B. Linux

Linux came out on top overall, with consistently high scores in the security, performance and kernel architecture categories. This is in line with the literature reviewed, which identified Linux as highly customisable, and equipped with advanced security tools such as SELinux and AppArmor [5][13]. Its low usability score is reflective of the well documented requirement for technical knowledge to configure and maintain the system effectively [13][15]. However, the high weighting given to security and kernel architecture in the Desktop/Enterprise profile was sufficient to put Linux at the top of the ranking.

C. macOS and Windows

macOS was second with a balance of performance, usability, and security scores. It has a controlled ecosystem with strong sandboxing and encryption mechanisms noted by Chen et al. [3] which positively contributed to their security score. Windows was third despite scoring the highest in usability among all tested systems. This is consistent with literature findings that Windows remains the most targeted operating system by malware due to its large user base [1][14] which directly lowered its security score and pulled its overall ranking below macOS. The small gap between macOS (0.5686) and Windows (0.5640) suggests that the two systems are closely matched for a general desktop profile and the ranking between them could be reversed depending on the weight given to usability versus security.

D. Mobile Operating Systems

iOS ranked fourth and Android fifth. Both systems received good ratings on usability and suitability for IoT, but lower scores on kernel architecture and real-time capability, which are heavily weighted in the Desktop/Enterprise profile. These findings are consistent with the literature which states that mobile operating systems are optimized for consumer experience and application ecosystems, rather than kernel-level robustness or deterministic performance [9][20]. iOS's ranking above Android is primarily due to its higher security score, reflecting its closed ecosystem and stricter application sandboxing policies.

E. Embedded and Real-Time Systems

Raspbian and FreeRTOS came sixth and seventh in the Desktop/Enterprise profile.This outcome does not reflect poor design but rather a mismatch between their strengths and the weight profile applied. FreeRTOS achieved the highest real-time capability score (10.0) and a strong IoT suitability score (9.0), making it the strongest candidate under an Embedded/Real-Time weight profile where those dimensions carry greater importance [2][8].Likewise, Raspbian is specifically designed for resource-constrained and educational computing contexts [10]. The lower performance and usability scores are to be expected considering the deployment context that it is targeted for. This exemplifies a key strength of the SMOCF framework: the same operating system can have substantially different rankings depending on the applied deployment profile, which provides a more realistic interpretation of the selection decisions in practice than a single fixed ranking.

F. Limitations of the framework

Although the SMOCF framework offers a reproducible and systematic basis for OS comparison, there are some limitations that need to be acknowledged. First, the dimension scores used in this study are based on the reviewed literature rather than experimental benchmarks, which adds a subjective element to the score assignment. Second, the weight profiles defined for each deployment context are based on general assumptions and may not be consistent with the priorities of all multiple evaluation dimensions including security, performance, kernel architecture, usability, IoT suitability, and real-time capability.

VII. Findings And Future Work

In this study, we carried out a comparative analysis of seven operating systems in six evaluation dimensions with the help of the SMOCF framework and TOPSIS method. The main findings are summarized as follows:

In the Desktop/Enterprise profile, Linux was the most balanced operating system with the highest closeness coefficient (0.7022) due to its strong security architecture, kernel robustness and performance [5][13].

Windows and macOS had similar results, macOS had a bit higher score due to better security mechanisms while Windows outperformed in usability [1][3][14].

Mobile operating systems iOS and Android have scored well with usability and ecosystem support but have scored less under the desktop profile due to weaker kernel architecture and real-time capability scores [9].

The real-time capability score of FreeRTOS was the best among all the systems that were tested, which confirms the appropriateness of FreeRTOS for embedded and IoT environments, although it scored low under the desktop profile [2][10].

The framework validated that there is no one-size-fits-all operating system for all deployment scenarios. When the weight profile changes the rankings change considerably, which highlights the importance of selecting an operating system based on the context.

A. Future Work

Future Research Directions Using live data from CVE databases for security, and standardized tools such as Phoronix Test Suite for performance [5][12] instead of scores derived from literature could improve the objectivity of the framework. Weight refinement: The subjectivity of the scoring model can be reduced by means of expert-based methods, e.g., Analytic Hierarchy Process (AHP), to determine the weights of the dimensions [12].

More extensive OS coverage: The comparison could be extended to cover more recent or emerging operating systems such as Chrome OS, Harmony OS and Ubuntu Core. Evaluation against real-world attack scenarios: A thorough evaluation of Windows, macOS, and Linux against real-world attack scenarios would be a useful complement to the quantitative framework proposed in this paper and would fill the main gap in the literature [1][15][16].

Mobile and cloud environments: The framework could be extended to evaluate mobile and cloud operating systems, considering the increasing importance of these platforms in today's computing environments [14].

VIII. Conclusion

In this research, a comparative analysis of different operating systems was conducted to evaluate their security mechanisms and effectiveness against modern cybersecurity threats. The study focused on Windows, macOS, Linux, Android, iOS, Raspberry Pi OS, and FreeRTOS using multiple evaluation dimensions including security, performance, kernel architecture, usability, IoT suitability, and real-time capability.

The proposed Structured Multi-Dimensional OS Comparison Framework (SMOCF) combined with the TOPSIS ranking method provided a systematic approach for evaluating and comparing operating systems across different computing environments. The findings showed that Linux achieved the highest overall ranking under the Desktop/Enterprise profile due to its strong security architecture, kernel robustness, and performance efficiency. macOS demonstrated strong protection mechanisms and system stability, while Windows maintained high usability and software compatibility despite facing greater exposure to malware attacks.

The study also highlighted that no single operating system is ideal for all deployment scenarios. Each operating system provides different strengths depending on the application environment, security requirements, and user needs. Real-time and embedded systems such as FreeRTOS and Raspberry Pi OS demonstrated strong capabilities in IoT and embedded environments, even though they ranked lower in desktop-focused evaluations.

Overall, this research contributes to a better understanding of operating system security mechanisms and provides a structured framework for comparing operating systems using unified evaluation criteria. The proposed framework can support users, organizations, and researchers in selecting suitable operating systems based on security and operational requirements. Future work may focus on integrating real-world benchmarking data, live vulnerability databases, and advanced attack simulation techniques to further improve the accuracy and effectiveness of operating system security evaluation.

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